

Two-body initial and boundary value problem

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- Initial value problem: propagation
- Boundary value problem: Lambert problem

Orbit propagation

Given an orbital state vector at one time, what is it at a future time?

$$\begin{bmatrix} \vec{r} \\ \dot{\vec{r}} \end{bmatrix} \rightarrow \begin{bmatrix} \vec{r}(t) \\ \dot{\vec{r}}(t) \end{bmatrix} \quad \vec{r}_0 = \vec{r}(t_0)$$

↑
epoch time

Method of Lagrange coefficients "the f and g functions"

Curtis section 2.11, but I'm going to use the change in eccentric anomaly rather than in true anomaly

$$\vec{r}(t) = f(t, t_0) \vec{r}_0 + g(t, t_0) \dot{\vec{r}}_0$$

$$\dot{\vec{r}}(t) = \dot{f}(t, t_0) \vec{r}_0 + \dot{g}(t, t_0) \dot{\vec{r}}_0$$

$$f(t_0, t_0) = 1 = \dot{g}(t_0, t_0)$$

$$\dot{f}(t_0, t_0) = 0 = g(t_0, t_0)$$

$$\vec{r}_0 \times \dot{\vec{r}}_0 = (f \vec{r}_0 + g \dot{\vec{r}}_0) \times \dot{\vec{r}}_0 = f(\vec{r}_0 \times \dot{\vec{r}}_0) = f \vec{h}$$

$$f = \frac{|\vec{r} \times \dot{\vec{r}}_0|}{h} = \frac{x \dot{y}_0 - \dot{x}_0 y}{h}$$

$$\dot{f} = \frac{\dot{x} \dot{y}_0 - \dot{x}_0 \dot{y}}{h}$$

$$\dot{\vec{r}}_0 \times \vec{r} = \dot{\vec{r}}_0 \times (f \vec{r}_0 + g \dot{\vec{r}}_0) = g \vec{h}$$

$$g = \frac{x_0 y - x y_0}{h}, \quad \dot{g} = \frac{x_0 \dot{y} - \dot{x}_0 y_0}{h}$$

The f, g in terms of Delta E

$$\cos E = e + \frac{r \cos \theta}{a} = e + \frac{x}{a}$$

$$a\sqrt{1-e^2} \sin E = r \sin \theta = y$$

$$x = a(\cos E - e)$$

$$y = a\sqrt{1-e^2} \sin E$$

$$\dot{x} = -a\dot{E} \sin E$$

$$\dot{y} = a\dot{E} \sqrt{1-e^2} \cos E$$

$$n = \frac{dM}{dt} = \dot{E} - e\dot{E} \cos E = \dot{E} \underbrace{(1 - e \cos E)}_{\frac{r}{a}} = \dot{E} \frac{r}{a}$$

$$\dot{E} = \frac{a_n}{r} = \frac{1}{r} \sqrt{\frac{\mu}{a}}$$

$$\begin{aligned} \dot{W} &= \dot{x} \dot{y} - \dot{x}_0 \dot{y}_0 = a(\cos E - e) a\dot{E} \sqrt{1-e^2} \cos E \\ &\quad + a\dot{E} \sin E \cdot a\sqrt{1-e^2} \sin E \\ &= a^2 \sqrt{1-e^2} \dot{E} [\cos E (\cos E - e) + \sin E \sin E] \end{aligned}$$

$$\cos(E - E_0) = \cos E \cos E_0 + \sin E \sin E_0$$

$$\begin{aligned} f &= \frac{a^2 \sqrt{1-e^2}}{h} \frac{1}{r_0} \sqrt{\frac{\mu}{a}} [\cos(\underbrace{E - E_0}_{\Delta E}) - e \cos E_0] \\ &= \frac{a}{r_0} [\cos \Delta E - (1 - \frac{r_0}{a})] \\ &= 1 - \frac{r_0}{a} (1 - \cos \Delta E) \end{aligned} \quad h = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)}$$

Lagrange coefficients

$$f = 1 - \frac{a}{r_{\bullet}} (1 - \cos \Delta E)$$

$$g = \Delta t - \sqrt{\frac{a^3}{\mu}} (\Delta E - \sin \Delta E)$$

$$\dot{f} = \frac{-\sqrt{\mu a} \sin \Delta E}{r r_{\bullet}}$$

$$\dot{g} = 1 - \frac{a}{r} (1 - \cos \Delta E).$$

Example 2.13

In universal variables

Universal anomaly χ Stumpff functions C, S

$$f = 1 - \frac{\chi^2}{r_0} C(\alpha \chi^2) \quad (3.69a)$$

$$g = \Delta t - \frac{1}{\sqrt{\mu}} \chi^3 S(\alpha \chi^2) \quad (3.69b)$$

$$\dot{f} = \frac{\sqrt{\mu}}{r_0} [\alpha \chi^3 S(\alpha \chi^2) - \chi] \quad (3.69c)$$

$$\dot{g} = 1 - \frac{\chi^2}{r} C(\alpha \chi^2) \quad (3.69d)$$

$$\alpha = \frac{1}{a} = \left(\frac{2}{r_{\bullet}} - \frac{v_{\bullet}^2}{\mu} \right)$$

Propagation with Lagrange coefficients

$$\begin{bmatrix} \mathbf{r}(t) \\ \dot{\mathbf{r}}(t) \end{bmatrix} = \begin{bmatrix} f(t, t_0) \mathbf{I} & g(t, t_0) \mathbf{I} \\ \dot{f}(t, t_0) \mathbf{I} & \dot{g}(t, t_0) \mathbf{I} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_0 \\ \dot{\mathbf{r}}_0 \end{bmatrix}$$

↘ 3x3 identity matrix

State transition matrix (STM)

The Jacobian matrix (matrix of partial derivatives) of the orbital state at the final time with respect to the state at the initial time.

Battin's method

Arora, Russell, Strange, Ottesen "Partial derivatives of the solution to the Lambert boundary value problem" JGCD, 2015, doi: 10.2514/1.G001030

Universal formulation

Battin's technique formulates the STM blocks

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

↑
A - ...

3x3 blocks

Lecture 10
10/1/2026

Initial position and velocity subscript 1
Final subscript 2

$$\alpha = \frac{2}{r_1} - \frac{v_1^2}{\mu}$$

$$\sigma_1 = \frac{\vec{r}_1^T \vec{v}_1}{\sqrt{\mu}}, \quad \sigma_2 = \frac{\vec{r}_2^T \vec{v}_2}{\sqrt{\mu}}$$

$$\chi = \alpha \sqrt{\mu} \underset{\substack{\uparrow \\ \text{time of flight}}}{\text{TOF}} + \sigma_2 - \sigma_1$$

$$U_1, \dots, U_5$$

$$U_1 = -\frac{r_2 r_1 f}{\sqrt{\mu}}, \quad U_2 = r_1 (1-f), \quad U_3 = \sqrt{\mu} (\text{TOF} - g)$$

$$U_4 = U_1 U_3 - \frac{1}{2} (U_2^2 - \alpha U_3^2)$$

$$U_5 = \frac{\frac{1}{6} \chi^3 - U_3}{\alpha}$$

$$\underline{\Phi}(t_2, t_1) = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{\partial \vec{r}_2}{\partial \vec{r}_1} & \frac{\partial \vec{r}_2}{\partial \vec{v}_1} \\ \frac{\partial \vec{v}_2}{\partial \vec{r}_1} & \frac{\partial \vec{v}_2}{\partial \vec{v}_1} \end{bmatrix}$$

$$\Phi(t_2, t_1) = \begin{bmatrix} C & D \end{bmatrix} = \begin{bmatrix} \frac{\partial \mathbf{r}_2}{\partial \mathbf{r}_1} & \frac{\partial \dot{\mathbf{r}}_2}{\partial \dot{\mathbf{r}}_1} \end{bmatrix}$$

$$\bar{C} = \frac{1}{\sqrt{\mu}} (3U_5 - \chi U_4 - \sqrt{\mu} \text{ToF } U_2)$$

1. Position-to-Position Submatrix (A)

$$A = \frac{r_2}{\mu} (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)(\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)^T + \frac{1}{r_1^3} \left[r_1(1-f) \mathbf{r}_2 \mathbf{r}_1^T + \bar{C} \dot{\mathbf{r}}_2 \mathbf{r}_1^T \right] + fI$$

2. Velocity-to-Position Submatrix (B)

$$B = \frac{r_1}{\mu} (1-f) \left[(\mathbf{r}_2 - \mathbf{r}_1) \dot{\mathbf{r}}_1^T - (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1) \mathbf{r}_1^T \right] + \frac{\bar{C}}{\mu} \dot{\mathbf{r}}_2 \dot{\mathbf{r}}_1^T + gI$$

3. Position-to-Velocity Submatrix (C)

$$C = -\frac{1}{r_1^2} (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1) \mathbf{r}_1^T - \frac{1}{r_2^2} \mathbf{r}_2 (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)^T + \dot{f} \left[I - \frac{1}{r_2^2} \mathbf{r}_2 \mathbf{r}_2^T + \frac{1}{\mu r_2} (\mathbf{r}_2 \dot{\mathbf{r}}_2^T - \dot{\mathbf{r}}_2 \mathbf{r}_2^T) \mathbf{r}_2 (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1) \right] - \frac{\mu \bar{C}}{r_1^3 r_2^3} \mathbf{r}_2 \mathbf{r}_1^T$$

4. Velocity-to-Velocity Submatrix (D)

$$D = \frac{r_1}{\mu} (\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)(\dot{\mathbf{r}}_2 - \dot{\mathbf{r}}_1)^T + \frac{1}{r_2^3} \left[r_1(1-f) \mathbf{r}_2 \mathbf{r}_1^T - \bar{C} \mathbf{r}_2 \dot{\mathbf{r}}_1^T \right] + \dot{g}I$$

3×3 identity matrix

Comparison of orbits

It is often useful to compare orbits by looking at the magnitude of the position differences